

TITLE: AI LECTURE COMPANION

PROBLEM: DESCRIPTION:

University students often struggle to listen and take notes simultaneously, causing them to miss important explanations, examples, formulas, and numerical steps.

The AI Lecture Companion automatically processes lecture content and generates organized notes and key information, allowing students to focus on understanding the lecture instead of manually taking notes.

PROBLEM IN AGENT'S CONTEXT

The agent needs to handle the problem of incomplete or unorganized lecture information.

that students may have after a university lecture. Students can miss important concepts, explanations, examples, formulas or numerical steps while trying to understand the teacher and take notes simultaneously.

The agent's main problem is to ensure that the important information from a lecture is available to the student in a complete and organized form.

SOLUTION:

The AI Lecture Companion provides an automatic note-making system.

* Working:

① The teacher's audio lecture is recorded.

- ② The agent converts the audio into text.
- ③ Board pictures processing.
- ④ Teachers slides and notes are processed.
- ⑤ The agent combines information from all sources.
- ⑥ Important concepts and examples are identified.
- ⑦ Additional reliable information can be retrieved.
- ⑧ The agent prepares complete easy-to-understand notes.
- ⑨ It includes examples, formulas, diagrams and solved numericals.
- ⑩ It generates practice questions exercises and answers.

⑪ The material is organized lecture-wise.

Therefore, during class, the student can concentrate on the teacher and understand the lecture, instead of continuously writing notes.

PROBLEM FORMULATION:

a) TEXTUAL:

1) INITIAL STATE:

The teacher is giving a lecture, and the student has to listen and manually take notes.

2) ACTIONS:

The agent:
→ Processes lecture audio

- Reads board pictures, slides and notes.
- Combine and understands the information.
- Generates lecture wise study material.

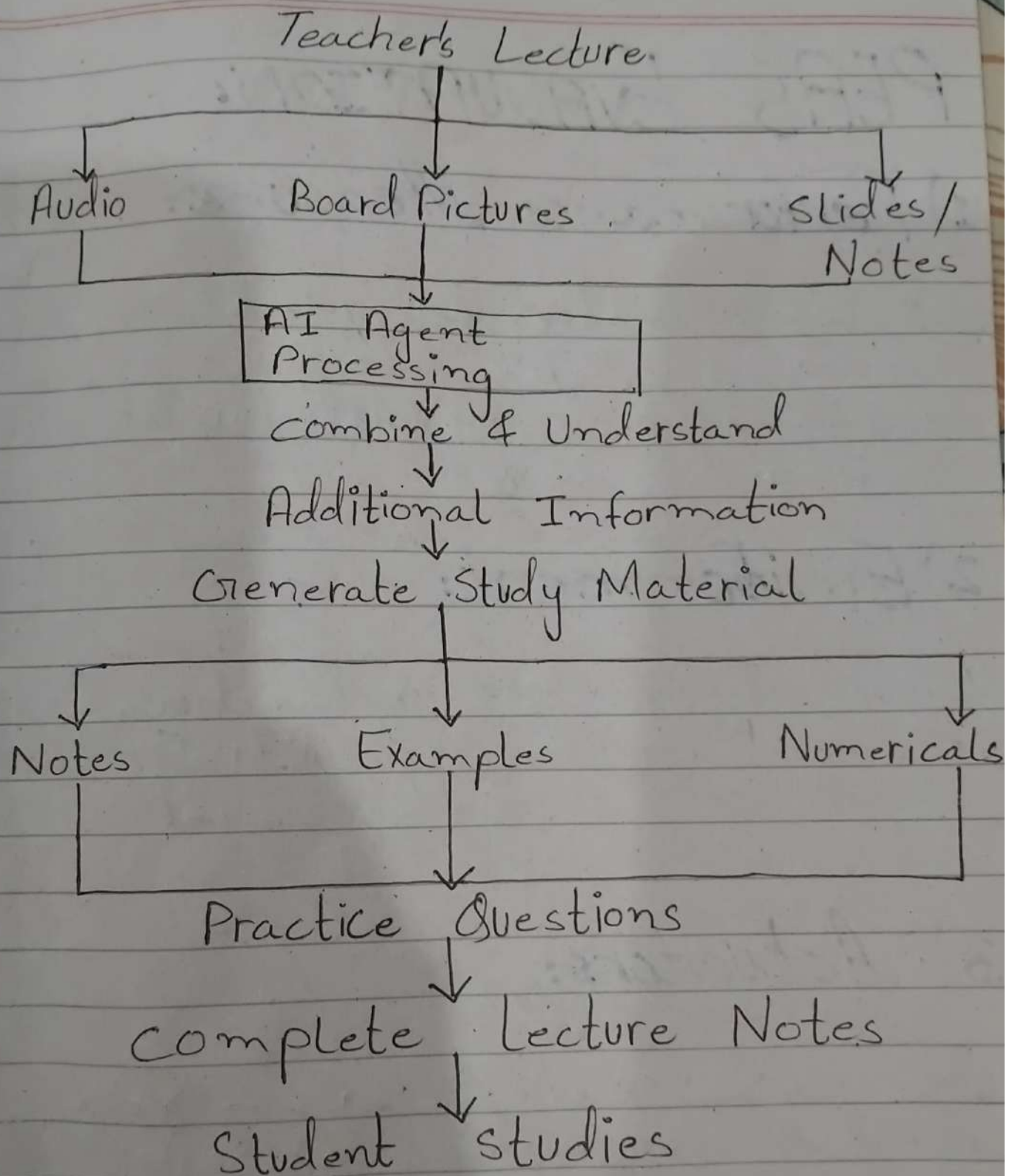
3) TRANSITION MODEL:

Lecture → Collect Material → Process information → Understand → Generate Notes → Final Study Material.

4) GOAL STATE:

The student receives complete and organized lecture notes without manually writing everything during class.

b) GRAPHICAL FORMULATION



PEAS EVALUATION:

1) Performance Measure:

- Accurate notes, complete lecture coverage, clear explanations, correct solutions, and time saved.

2) Environment:

University classroom, teacher, lectures, board, slides, notes and online educational sources.

3) Actuators:

Generates notes, examples, solutions, questions, exercises and revision material

4) Sensors:

Microphone/audio, camera/

board pictures, slides, notes,
and external information
sources.

ENVIRONMENT PROPERTIES:

Property	Type	Reason
→ Observability	Partially Observable	The agent may not capture every spoken or written detail perfectly.
→ Determinism	Stochastic	Speech recognition, image processing and AI generation may produce uncertain results.
→ Episodes	Sequential	New lectures can depend on concepts taught in previous lectures.
→ Change	Dynamic	New lectures and

materials are continuously added.

→ Values	Discrete & continuous	Text, questions and topics are discrete, while audio is continuous.
→ Agents	Multi-Agent	Teacher, student and AI agent are involved in the learning environment.